

Knowvo Parent – Project Summary

Knowvo Parent is a React Native-based multi-tenant School ERP application that connects parents and students with their school ecosystem. A single codebase powers 50+ white-label applications for different institutions, where branding, module access, and payment gateway configurations are dynamically controlled based on school-specific settings. The application provides academic, communication, transportation, fee management, learning, and e-commerce services through a unified platform.

Module Summaries

Dashboard

The dashboard is the home screen after login. It displays the student profile, a dynamic feature grid, and summary cards for attendance, fees, performance, and upcoming classes. Available modules are shown based on school configuration, and announcements are displayed through a banner carousel.

Attendance

The attendance module provides a calendar view of attendance history with indicators for present, absent, and holidays. Parents can view attendance percentage, total working days, present days, and absent days. Data can also be viewed academic year-wise.

Performance

The performance module provides academic analytics and report cards. Parents can view subject-wise and exam-wise results and compare student scores with campus average and maximum scores through graphical representations and detailed mark sheets.

Fees & Payments

This module manages fee dues and online payments. It displays detailed fee components and supports multiple payment gateways such as Razorpay, PhonePe, and Grayquest. Parents can make full payments, installment payments, or opt for EMI plans. Payment history and receipts are also available.

Digital Diary

The digital diary acts as a communication platform between schools and parents. It includes homework, circulars, activity updates, wellness notes, and announcements. Entries can be searched, filtered by date range, and acknowledged with remarks.

Notifications

The notification module serves as a centralized inbox for application events. Notifications include fee reminders, attendance alerts, diary entries, bus updates, and order status changes. Notifications are delivered using Firebase Cloud Messaging and support unread tracking and filtering.

Learn 2.0 (StudyHub)

StudyHub is the learning platform of the application. It combines native functionality with WebView-based content to provide videos, practice tests, doubt-solving, and premium courses. Learning materials can be downloaded and accessed offline, with support for ZIP extraction and USB-based content transfer.

Smart Classes (WAVE)

Smart Classes provides access to live virtual classrooms through Vedantu WAVE integration. Sessions are categorized into LIVE, UPCOMING, COMPLETED, and CANCELLED states. Students can browse schedules using a calendar interface and join live sessions before the scheduled start time.

Transportation

The transportation module enables real-time school bus tracking. Bus locations are streamed through Firebase Realtime Database and displayed on Google Maps. Route paths, bus stop indicators, driver information, and location freshness status help parents monitor bus movement.

E-Store

The e-Store module allows schools to sell books, uniforms, and other supplies. Parents can browse products, apply filters, add items to the cart, complete payments, view order history, and raise complaints related to orders.

Profile

The profile module manages student information and supports multiple children under a single parent account. Users can update profile details, upload profile pictures, and switch between children. Module-specific caches are cleared during child switching to avoid stale data from another student.

Knowvo Launcher

Knowvo Launcher is a React Native and TypeScript based Android Kiosk application developed specifically for schools to create a distraction-free learning environment. The app becomes the default home screen, so whenever the Android box is turned on, Knowvo Launcher automatically starts and students cannot close or exit it. Only approved educational apps are visible, and students cannot access other applications, system settings, or perform unwanted actions. Even if the teacher leaves the classroom, students can continue learning without getting distracted by games, YouTube, or other apps. Any admin actions, such as adding or removing apps, are protected by a password so only authorized staff can make changes. Using Android Device Owner and Kiosk Mode, the application takes control of the device at the system level and converts shared classroom devices into secure learning devices, ensuring students stay focused on education and preventing misuse of school devices.